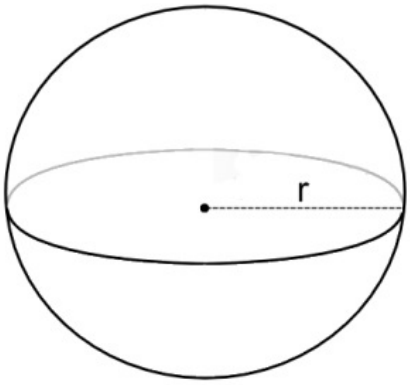
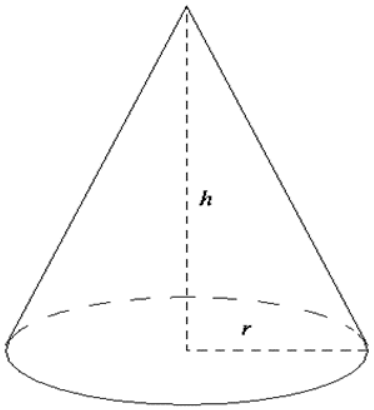
1. Write a java program to calculate the volume of Sphere and Cone, requirements are given below.

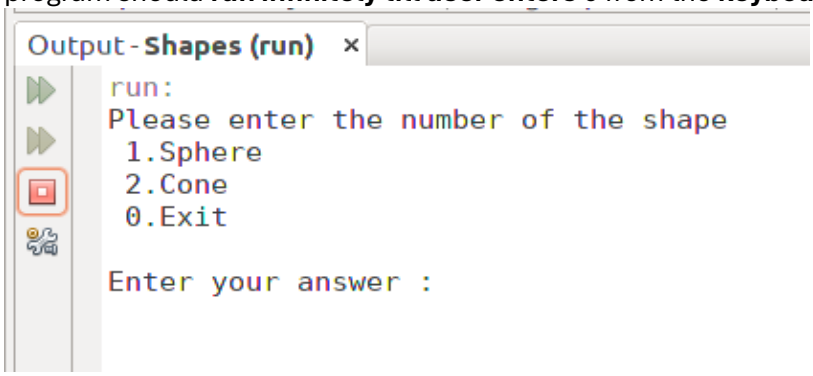




Sphere Volume = $(4/3)\ \pi r^3$ Cone Volume = $(1/3)\ \pi r^2 h$

a. You should get all **radius(r)** and **height(h)** values as **double** values as **user inputs** for each object from **keyboard**.

b. You should prompt the following screen at the beginning of the program and program should **run infinitely till user enters 0** from the **keyboard**.



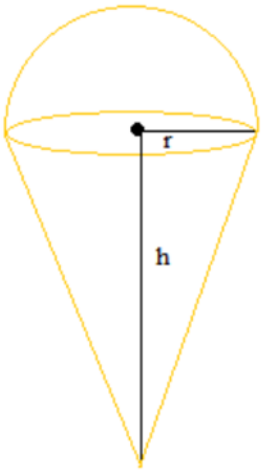
c. In your program you must write **two separate methods** as given below.

double getSphereVolume(double radius);
double getConeVolume(double radius, double height);

(Hint: use Math.PI as π value for calculations)

You can refer to "**Annex A**" for the complete flow of the program.

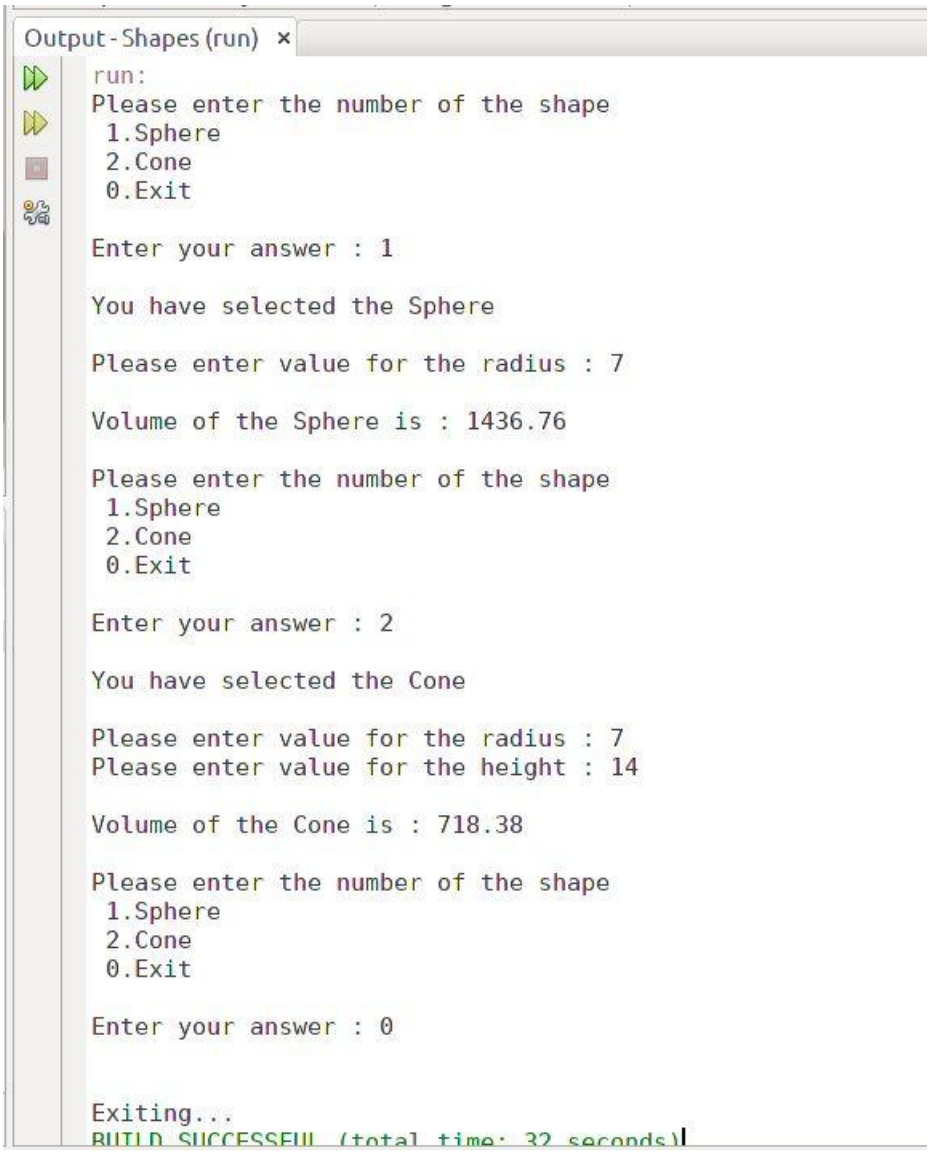
d. **Modify** your program to calculate the volume of the following shape by adding an additional item to your menu called "**MyShape**", and a method called "**getMyShapeVolume(double radius)**" to your program.

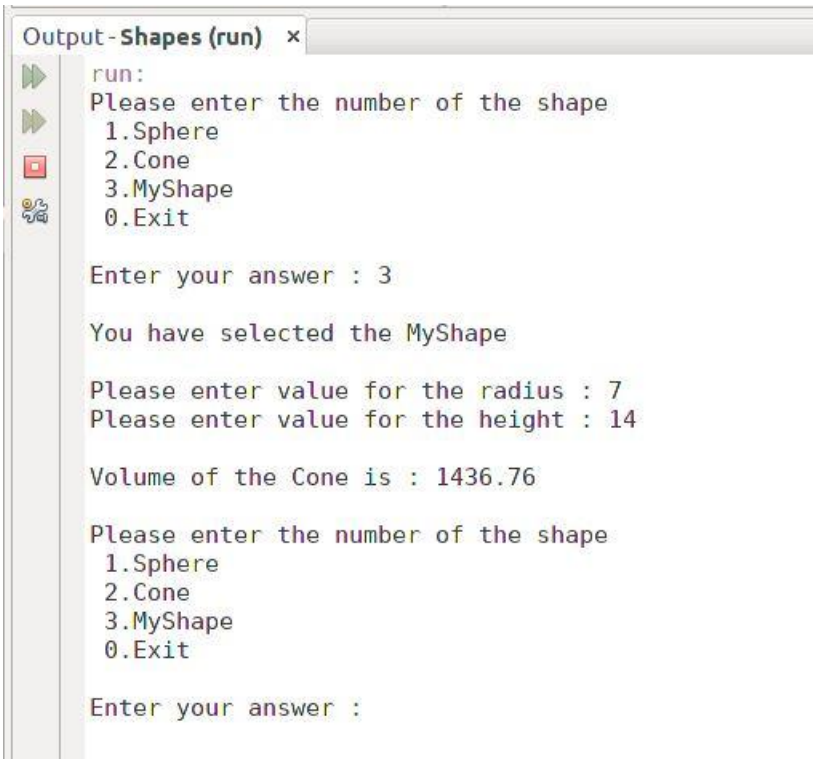


You can refer to "**Annex B**" for the complete flow of the modified program.

Inorder to complete above program you need to use a Driver class named "ShapeDemo" and 03 distinct class to represent 03 shapes "Sphere", "Cone" and "MyShape"

Annex A


Output - Shapes (run)
run:
Please enter the number of the shape
1.Sphere
2.Cone
0.Exit
Enter your answer : 1
You have selected the Sphere
Please enter value for the radius : 7
Volume of the Sphere is : 1436.76
Please enter the number of the shape
1.Sphere
2.Cone
0.Exit
Enter your answer : 2
You have selected the Cone
Please enter value for the radius : 7
Please enter value for the height : 14
Volume of the Cone is : 718.38
Please enter the number of the shape
1.Sphere
2.Cone
0.Exit
Enter your answer : 0
Exiting...
BUILD SUCCESSFUL (total time: 32 seconds)


Output - Shapes (run)
run:
Please enter the number of the shape
1.Sphere
2.Cone
3.MyShape
0.Exit
Enter your answer : 3
You have selected the MyShape
Please enter value for the radius : 7
Please enter value for the height : 14
Volume of the Cone is : 1436.76
Please enter the number of the shape
1.Sphere
2.Cone
3.MyShape
0.Exit
Enter your answer :

Annex B